

$3 \rightarrow 2 \cdot 3 \rightarrow 4 \rightarrow 2$

Distance

$D_4 =$

	1	2	3	4	5
1	-	3	10	8	12
2	3	-	11	5	9
3	10	11	-	6	10
4	8	5	6	-	4
5	12	9	10	4	-

 $k=5$ $S_4 =$

	1	2	3	4	5
1	-	2	3	2	4
2	1	-	4	4	4
3	1	4	-	4	4
4	2	2	3	-	5
5	4	4	4	4	-

No change.

$$k=6 = n+1$$

Shortest route between 1-5:

Shortest distance: 12.

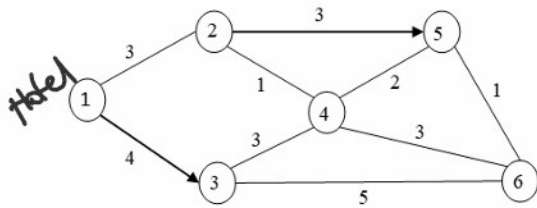
Shortest route =

 $S_{15} = 4$ indirect.
 $1 \rightarrow 4 \rightarrow 5$
 $S_{14} = 2$ $S_{45} = 5$
 indirect direct.

 $1 \rightarrow 2 \rightarrow 4 \rightarrow 5$ Shortest route.
 $S_{12} = 2$ $S_{24} = 4$

Shortest route between : 3 - 2.

Q2) (30 p.) The Hylton Hotel has a limousine that transports guests to various business and tourist locations around the city. The following network indicates the different routes the limousine could follow from the hotel at node 1 to the five locations (nodes 2 through 6): The values on each arc in the network are the distances, in miles, between the locations. By implementing an algorithm, determine all the possible shortest routes from the hotel to each of the five destinations and indicate the distance for each route.



-	3	4	∞	∞	∞
3	-	∞	1	3	∞
8	8	-	3	∞	5
8	1	3	-	2	3
8	8	8	2	-	1
8	8	5	3	1	-

$k=1$

-	2	3	4	5	6
1	-	3	4	5	6
1	2	-	4	5	6
1	2	3	-	5	6
1	2	3	4	-	6
1	2	3	4	5	-

-	3	4	∞	∞	∞
3	-	7	1	3	∞
8	8	-	3	∞	5
8	1	3	-	2	3
8	8	8	2	-	1
8	8	5	3	1	-

$k=2$

-	2	3	4	5	6
1	-	1	4	5	6
1	2	-	4	5	6
1	2	3	-	5	6
1	2	3	4	-	6
1	2	3	4	5	-

-	3	4	4	6	∞
3	-	7	1	3	∞
8	8	-	3	∞	5
4	1	3	-	2	3
8	8	8	2	-	1
8	8	5	3	1	-

$k=3$

-	2	3	2	2	6
1	-	1	4	5	6
1	2	-	4	5	6
2	2	3	-	5	6
1	2	3	4	-	6
1	2	3	4	5	-

-	3	4	4	6	9
3	-	7	1	3	12
8	8	-	3	∞	5
4	1	3	-	2	3
8	8	8	2	-	1
8	8	5	3	1	-

$k=4$

-	2	3	2	2	3
1	-	1	4	5	3
1	2	-	4	5	6
2	2	3	-	5	6
1	2	3	4	-	6
1	2	3	4	5	-

No improvement is caught in this iteration.

-	3	4	4	6	7
3	-	4	1	3	4
7	4	-	3	5	5
4	1	3	-	2	3
6	3	5	2	-	1
7	4	5	3	1	-

$k=5$
 $k=6$

-	2	3	2	2	4
1	-	4	4	5	4
4	4	-	4	4	6
2	2	3	-	5	6
4	4	4	4	-	6
4	4	3	4	5	-

No change.

this iteration.

No change.

$k=7$. STOP
(n+1).

S.R. $1 \rightarrow 6$ 7 miles.

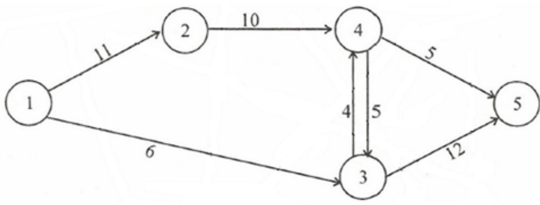
$$S_{16} = 4 \Rightarrow 1 \rightarrow 4 \rightarrow 6$$

$S_{14} = 2 \quad S_{46} = 6.$

$1 \rightarrow 2 \rightarrow 4 \rightarrow 6$

$S_{12} = 2$

Maximal number
of cuts?



$$N = \{1, 2, 3, 4, 5\}$$



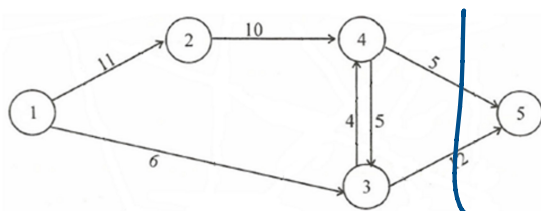
$$n \rightarrow 2^n.$$

$$\begin{matrix} S & \{2, 3, 4\} & \bar{S} \\ \downarrow & \downarrow & \downarrow \\ \{1\} & ? & \{5\} \end{matrix}$$

2^3

$$\begin{matrix} n-2 \\ 2 \end{matrix}$$

$n \rightarrow$ maximal number of
cuts



$$11 + 5 + 12$$

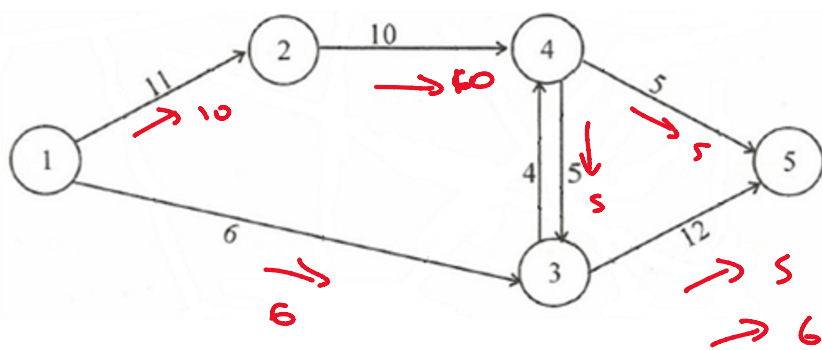
$\rightarrow \phi$ $c.$ 12

$c = \{1, 3\}$ $c = \{2, 3, 4, 5\}$ $c_1: 17$

$\rightarrow 9$ $c.$ 12
 $S = \{1\}$ $\bar{S} = \{2, 3, 4, 5\}$ $C_1 = 17$
 $S = \{1, 2\} \rightarrow \{2\}$ $\bar{S} = \{3, 4, 5\}$ $C_2 = 16$
 $S = \{1, 3\} \rightarrow \{3\}$ $\bar{S} = \{2, 4, 5\}$ $C_3 = 11 + 12 + 4 = 27$
 $S = \{1, 4\} \rightarrow \{4\}$ $\bar{S} = \{2, 3, 5\}$ $C_4 = 27$
 $S = \{1, 2, 3\}$ $\bar{S} = \{4, 5\}$ $C_5 = 26$
 $S = \{1, 2, 4\}$ $\bar{S} = \{3, 5\}$ $C_6 = 16$
 $S = \{1, 3, 4\}$ $\bar{S} = \{2, 5\}$ $C_7 = 28$
 $S = \{1, 2, 3, 4\}$ $\bar{S} = \{5\}$ $C_8 = 17$

$\min_{\text{all cuts}} C_i = \text{maximal flow.}$

16



$S: 1 \rightarrow 2 \rightarrow 4 \rightarrow 5$

$S: 1 \rightarrow 2 \rightarrow 4 \rightarrow 3 \rightarrow 5$

$+ 6: 1 \rightarrow 3 \rightarrow 5$

 16.